SOCIOL 723 Social Statistics II
Estimation, Causal Inference, and Machine Learning

Time and Location

Fall 2026
Tuesdays (Lecture) and Thursdays (Lab), 10:05AM–12:35PM
Erwin Square Plaza Room 753

Instructor Information

Professor	Wenhao Jiang
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Office Hours	Wednesdays 2–4PM or by appointment

Course Overview and Prerequisites

This is the second course in the graduate statistics sequence for sociology doctoral students. *Social Statistics I*, using the model-comparison approach of Correll et al., rigorously developed regression, ANOVA, and a first look at logistic and Poisson regression in `R`. Building on that foundation, this course extends into more advanced quantitative methodology, organized around three pillars: estimation, causal inference, and machine learning. We begin by revisiting ordinary least squares (OLS), recasting the model-comparison framework in matrix form and attending carefully to its statistical properties, its interpretation as a tool for causal adjustment, and inference under realistic assumptions, including heteroskedasticity- and cluster-robust standard errors. We then develop maximum likelihood estimation (MLE) as the general engine behind most models sociologists use, grounding the logistic and Poisson models introduced in *Social Statistics I* in likelihood theory and extending them, within the generalized linear model framework, to multinomial, ordered, and other limited dependent variables.

The core of the course turns to causal inference, the central methodological concern of contemporary empirical social science. We introduce the potential outcomes framework and directed acyclic graphs (DAGs) as complementary languages for defining and identifying causal effects, then work through the standard research designs: selection on observables (matching, propensity scores, and weighting), instrumental variables (IV), regression discontinuity designs (RDD), difference-in-differences (DID) with panel data, and synthetic control. The course closes by introducing machine learning for prediction and its integration with causal inference, including double/debiased machine learning and the estimation of heterogeneous treatment effects.

Each week pairs a Tuesday lecture that develops the theory with a Thursday lab that implements the methods on real data in `R`. By the end of the course, students should be able to read and critically evaluate the quantitative sociological literature that uses these methods, and to apply them competently in their own research. The course assumes working knowledge of the general linear model and statistical inference at the level of *Social Statistics I*. Familiarity with matrix algebra and calculus is helpful but not required; the necessary tools are reviewed in the first two labs. All coursework is in `R`.

Expectations

Reading and participation (10%):
You are expected to complete the required readings before the class in which they are discussed and to participate actively in both lecture and lab. If a reading is difficult, come prepared to discuss what you did and did not follow.

Problem sets (30%):
There will be 5 problem sets, assigned roughly bi-weekly. They reinforce the lecture material and typically require `R` coding building on that week's lab. Problem sets are released on Tuesdays after class and are generally due two weeks later by Monday, 11:59PM (see the schedule for exact dates).

In-Class Midterm Exam (25%):
There will be an in-class midterm on **Thursday, October 15**, covering the estimation half of the course (OLS and MLE) and the foundations of causal inference through instrumental variables (Weeks 1–7). The exam is open book and open note but timed. You will not be required to write any code.

Final Exam (35%):
There will be a cumulative final exam during the University examination period in December; the Registrar sets the exact date and time. It covers the whole course, but weighted towards **Weeks 9–13**, since the midterm already examined Weeks 1–7. Like the midterm, it is open book and open note but timed, and you will not be required to write any code. What is examined is the material developed in scalar form, together with what each design assumes and what it estimates; frames marked with a star in the slides are for understanding and will not be examined.

Software

This class uses `R` throughout, and prior experience at the level of *Social Statistics I* is assumed. Please install a recent version of `R` and `RStudio` before the first lab, and use `RMarkdown` or `Quarto` for your assignments to support reproducibility.

Use of Artificial Intelligence (AI)

I encourage you to use AI tools to support your statistical learning, particularly for working through steps in a derivation and for debugging `R` code. AI assistance is permitted on problem sets, provided you understand every step of what you submit and write it up in your own words.

AI is **not** permitted during either exam. That is deliberate, and it is why the exams carry the weight they do: they are the only points in the course where what is being assessed is unambiguously *your own* understanding.

Course Materials

We will draw regularly on three texts, two of which are free online.

- [Effect] Huntington-Klein, Nick. 2022. *The Effect: An Introduction to Research Design and Causality*. CRC Press. (online)
- [MHE] Angrist, Joshua D. and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press. (online)
- [CCI] Morgan, Stephen L. and Christopher Winship. 2015. *Counterfactuals and Causal Inference: Methods and Principles for Social Research*, 2nd ed. Cambridge University Press.

The Effect and the Chernozhukov et al. text are living online books whose chapter numbers shift between editions; chapter titles are given where the number alone might mislead. Within each week, the readings are listed *in the order you should read them*: the first builds the intuition, and those that follow supply the rigour. *Further reading* is there for depth and for your own research; it is not examined.

The remaining texts are used only for particular weeks and are freely available or optional rather than required purchases:

- Ward, Michael D. and John S. Ahlquist. 2018. *Maximum Likelihood for Social Science: Strategies for Analysis*. Cambridge University Press. (maximum likelihood, written for social scientists)
- [ISL] James, Gareth et al. 2021. *An Introduction to Statistical Learning: with Applications in R*, 2nd ed. Springer. (online)
- [Pawitan] Pawitan, Yudi. 2013. *In All Likelihood: Statistical Modelling and Inference Using Likelihood*. Oxford University Press.
- [Mixtape] Cunningham, Scott. 2021. *Causal Inference: The Mixtape*. Yale University Press. (online)
- [CML] Chernozhukov, Victor et al. 2025. *Applied Causal Inference Powered by ML and AI*. (online)

For review of prerequisite material, students may consult the text used in *Social Statistics I*: Correll, Josh, Abigail M. Folberg, Charles M. Judd, Gary H. McClelland, and Carey S. Ryan, *Data Analysis: A Model Comparison Approach to Regression, ANOVA, and Beyond* (Routledge), together with its `R` companion, Stephen Vaisey's *DAMCA*.

Schedule

The schedule is tentative as of August 13, 2026. A detailed list of topics and readings, by week, appears below. Tuesday sessions are lectures and Thursday sessions are `R` labs. You are responsible for completing the *required readings* before the Tuesday lecture.

Week	**Dates**	**Topic**	**Problem sets** Assign	Due
1	Aug 25/27	The Linear Model and OLS		
2	Sep 1/3	Regression Inference and Robust Standard Errors	1	
3	Sep 8/10	Maximum Likelihood: Theory		
4	Sep 15/17	Maximum Likelihood: Applications	2	1
5	Sep 22/24	Causal Inference: Potential Outcomes and DAGs		
6	Sep 29/Oct 1	Matching, Propensity Scores, and Weighting	3	2
7	Oct 6/8	Instrumental Variables		
8	Oct 13/15	*Fall break* (Tues) and *In-class Midterm* (Thurs)		3
9	Oct 20/22	Regression Discontinuity Designs	4	
10	Oct 27/29	Panel Data and Difference-in-Differences		
11	Nov 3/5	Synthetic Control and Its Extensions	5	4
12	Nov 10/12	Machine Learning: Prediction and Regularization		
13	Nov 17/19	Causal Machine Learning		5
14	Nov 24	Synthesis and Exam Review		
	December	*Cumulative final exam* (examination period)		

Tuesday sessions are lectures; Thursday sessions are hands-on `R` labs applying the week's methods. Fall break is Oct 9–14 (no Tuesday class Oct 13); the Thanksgiving recess begins Nov 24, so there is no Thursday lab on Nov 26.

Week 1 (Aug 25/27): The Linear Model and OLS

Lecture: The general linear model in scalar and matrix form, connecting the model-comparison framework of *Social Statistics I* to matrix notation; the OLS estimator and its geometry; the Frisch–Waugh–Lovell theorem, omitted-variable bias, and regression as a tool for adjustment. *Lab*: `R` and `RStudio` refresher; a review of the matrix algebra used in the linear model; fitting and interpreting OLS.

- *Reading*: Effect, Chapter 13; MHE, §3.1–3.2.
- *Further reading*: CCI, Chapter 6; ISL, Chapter 3.

Week 2 (Sep 1/3): Regression Inference and Robust Standard Errors

Lecture: Sampling distribution of OLS and hypothesis testing; heteroskedasticity; robust and cluster-robust standard errors; the bootstrap; what robust standard errors can and cannot fix. *Lab*: a review of the calculus used in estimation and optimization; robust and clustered inference and the bootstrap in `R`.

- *Reading*: Effect, Chapter 15; MHE, Chapter 8.

- *Further reading*: Freedman, David A. 2006. "On the So-Called 'Huber Sandwich Estimator' and 'Robust Standard Errors'." *The American Statistician* 60(4), 299–302; Bertrand, Marianne, Esther Duflo, and Sendhil Mullainathan. 2004. "How Much Should We Trust Differences-in-Differences Estimates?" *Quarterly Journal of Economics* 119(1), 249–275.

Week 3 (Sep 8/10): Maximum Likelihood: Theory

Lecture: The likelihood and log-likelihood; the score function, Hessian, and Fisher information; identification and numerical maximization; the asymptotic properties of the MLE (consistency, asymptotic normality, and efficiency) under regularity conditions; and likelihood-ratio, Wald, and score tests. *Lab*: coding a likelihood and optimizing it numerically in `R`.

- *Reading*: Ward and Ahlquist, Chapters 1–2 and 4.
- *Further reading*: Pawitan, Chapters 2–4.

Week 4 (Sep 15/17): Maximum Likelihood: Applications

Lecture: The generalized linear model framework (linear predictor, link, and likelihood); binary outcomes with logit and probit, now grounded in likelihood theory; interpretation via marginal effects, predicted probabilities, and interaction terms; multinomial and ordered models; count models (Poisson and negative binomial). *Lab*: estimating and interpreting limited-dependent-variable models in `R`.

- *Reading*: ISL, §4.1–4.3; Hanmer, Michael J. and Kerem Ozan Kalkan. 2013. "Behind the Curve: Clarifying the Best Approach to Calculating Predicted Probabilities and Marginal Effects from Limited Dependent Variable Models." *American Journal of Political Science* 57(1), 263–277.
- *Further reading*: Ward and Ahlquist, Chapter 3; Mize, Trenton D. 2019. "Best Practices for Estimating, Interpreting, and Presenting Nonlinear Interaction Effects." *Sociological Science* 6, 81–117; Zorn, Christopher. 2005. "A Solution to Separation in Binary Response Models." *Political Analysis* 13(2), 157–170.

Week 5 (Sep 22/24): Causal Inference: Potential Outcomes and DAGs

Lecture: The potential outcomes framework; SUTVA, ignorability, and identification; average treatment effects; directed acyclic graphs, back-door and front-door criteria, and collider bias; defining a well-posed estimand. *Lab*: simulating confounding and selection; drawing and reading DAGs in `R` (`dagitty`).

- *Reading*: Effect, Chapters 6 and 8; Lundberg, Ian, Rebecca Johnson, and Brandon M. Stewart. 2021. "What Is Your Estimand? Defining the Target Quantity Connects Statistical Evidence to Theory." *American Sociological Review* 86(3), 532–565; Elwert, Felix and Christopher Winship. 2014. "Endogenous Selection Bias: The Problem of Conditioning on a Collider Variable." *Annual Review of Sociology* 40, 31–53.
- *Further reading*: MHE, Chapter 2; CCI, Chapters 1–3.

Week 6 (Sep 29/Oct 1): Matching, Propensity Scores, and Weighting

Lecture: Selection on observables; exact, nearest-neighbor, and coarsened matching; the propensity score; inverse-probability weighting; doubly robust estimation; and what each method assumes when it fails. *Lab*: matching and weighting workflows in `R` (`MatchIt`, `WeightIt`).

- *Reading*: Effect, Chapter 14; CCI, Chapter 5; Stuart, Elizabeth A. 2010. "Matching Methods for Causal Inference: A Review and a Look Forward." *Statistical Science* 25(1), 1–21.
- *Further reading*: CCI, Chapters 4, 7, and 8; King, Gary and Richard Nielsen. 2019. "Why Propensity Scores Should Not Be Used for Matching." *Political Analysis* 27(4), 435–454.

Week 7 (Oct 6/8): Instrumental Variables

Lecture: Endogeneity and the IV estimator; two-stage least squares; the LATE framework, monotonicity, and compliers; weak instruments and sensitivity. *Lab*: 2SLS and diagnostics in `R` (`fixest/ivreg`).

- *Reading*: Effect, Chapter 19; MHE, §4.1 and §4.4 (the latter is where LATE and the compliance types live); Angrist, Joshua D., Guido W. Imbens, and Donald B. Rubin. 1996. "Identification of Causal Effects Using Instrumental Variables." *Journal of the American Statistical Association* 91(434), 444–455.
- *Further reading*: CCI, Chapter 9; Lee, David S., Justin McCrary, Marcelo J. Moreira, and Jack Porter. 2022. "Valid t-ratio Inference for IV." *American Economic Review* 112(10), 3260–3290.

Week 8 (Oct 13/15): Fall Break / In-Class Midterm

Tuesday, Oct 13: no class (Fall break, Oct 9–14). *Thursday, Oct 15*: in-class midterm covering Weeks 1–7.

Week 9 (Oct 20/22): Regression Discontinuity Designs

Lecture: Sharp and fuzzy RD; identification at the cutoff; local linear estimation, bandwidth choice, and kernel weighting; validity, sorting, and placebo checks. *Lab*: RD estimation and diagnostics in `R` (`rdrobust`).

- *Reading*: Effect, Chapter 20; Cattaneo, Matias D. and Rocío Titiunik. 2022. "Regression Discontinuity Designs." *Annual Review of Economics* 14, 821–851.
- *Further reading*: MHE, Chapter 6; McCrary, Justin. 2008. "Manipulation of the Running Variable in the Regression Discontinuity Design: A Density Test." *Journal of Econometrics* 142(2), 698–714.

Week 10 (Oct 27/29): Panel Data and Difference-in-Differences

Lecture: Panel data and unobserved heterogeneity; pooled OLS, first-difference, and fixed-effects (within) and random-effects estimators; two-way (unit and time) fixed effects (TWFE); the canonical 2×2 DID and the parallel-trends assumption; DID and event studies as special cases of TWFE; problems with TWFE under staggered adoption and recent robust estimators. *Lab*: panel and DID estimation in `R` (`plm`, `fixest`, `did`).

- *Reading*: Effect, Chapters 16 and 18; Roth, Jonathan, Pedro H.C. Sant'Anna, Alyssa Bilinski, and John Poe. 2023. "What's Trending in Difference-in-Differences? A Synthesis of the Recent Econometrics Literature." *Journal of Econometrics* 235(2), 2218–2244.
- *Further reading*: MHE, Chapter 5; CCI, Chapter 11; Goodman-Bacon, Andrew. 2021. "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics* 225(2), 254–277.

Week 11 (Nov 3/5): Synthetic Control and Its Extensions

Lecture: Comparative case studies; the synthetic control estimator; inference with few treated units; augmented and generalized synthetic control, and synthetic difference-in-differences. *Lab*: synthetic control in `R` (`Synth`, `gsynth`).

- *Reading*: Abadie, Alberto. 2021. "Using Synthetic Controls: Feasibility, Data Requirements, and Methodological Aspects." *Journal of Economic Literature* 59(2), 391–425.
- *Further reading*: Abadie, Alberto, Alexis Diamond, and Jens Hainmueller. 2010. "Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program." *Journal of the American Statistical Association* 105(490), 493–505; Xu, Yiqing. 2017. "Generalized Synthetic Control Method: Causal Inference with Interactive Fixed Effects Models." *Political Analysis* 25(1), 57–76.

Week 12 (Nov 10/12): Machine Learning: Prediction and Regularization

Lecture: Prediction versus inference; the bias–variance tradeoff; cross-validation; ridge, lasso, and elastic net; trees, random forests, and boosting. *Lab*: regularized regression and tree ensembles in `R` (`glmnet`, `ranger`).

- *Reading*: ISL, §2.1–2.2, 5.1, 6.2, and 8.1–8.2; Molina, Mario and Filiz Garip. 2019. "Machine Learning for Sociology." *Annual Review of Sociology* 45(1), 27–45.
- *Further reading*: Mullainathan, Sendhil and Jann Spiess. 2017. "Machine Learning: An Applied Econometric Approach." *Journal of Economic Perspectives* 31(2), 87–106.

Week 13 (Nov 17/19): Causal Machine Learning

Lecture: Why naive ML is biased for causal parameters; Neyman orthogonality and double/debiased machine learning (DML); partialling-out and cross-fitting; heterogeneous treatment effects and causal forests. *Lab*: DML and heterogeneous effects in `R` (`DoubleML`, `grf`).

- *Reading*: Athey, Susan and Guido W. Imbens. 2017. "The State of Applied Econometrics: Causality and Policy Evaluation." *Journal of Economic Perspectives* 31(2), 3–32; CML, Chapter 4.

- *Further reading*: CML, Chapter 10; Wager, Stefan and Susan Athey. 2018. "Estimation and Inference of Heterogeneous Treatment Effects Using Random Forests." *Journal of the American Statistical Association* 113(523), 1228–1242.

Week 14 (Nov 24): Synthesis and Exam Review

Tuesday, Nov 24: a synthesis of how estimation, causal identification, and machine learning fit together, followed by a review for the final exam: what each design assumes, what it estimates, and how to choose between them. *Thursday, Nov 26*: no class (Thanksgiving recess). The **final exam** is held in the December examination period.